

Combined Long Short-Term Memory based Network employing wavelet coefficients for MI-EEG recognition

Mingai Li, Meng Zhang, Xinyong Luo and Jinfu Yang

*College of Electronic Information & Control Engineering
Beijing University of Technology
Beijing 100124, China*

limingai@bjut.edu.cn

Abstract – Motor Imagery Electroencephalography (MI-EEG) plays an important role in brain computer interface (BCI) based rehabilitation robot, and its recognition is the key problem. The Discrete Wavelet Transform (DWT) has been applied to extract the time-frequency features of MI-EEG. However, the existing EEG classifiers, such as support vector machine (SVM), linear discriminant analysis (LDA) and BP network, did not make full use of the time sequence information in time-frequency features, the resulting recognition performance were not very ideal. In this paper, a Long Short-Term Memory (LSTM) based recurrent Neural Network (RNN) is integrated with Discrete Wavelet Transform (DWT) to yield a novel recognition method, denoted as DWT-LSTM. DWT is applied to analyze the each channel of MI-EEG and extract its effective wavelet coefficients, representing the time-frequency features. Then a LSTM based RNN is used as a classifier for the pattern recognition of observed MI-EEG data. Experiments are conducted on a publicly available dataset, and the 5-fold cross validation experimental results show that DWT-LSTM yields relatively higher classification accuracies compared to the existing approaches. This is helpful for the further research and application of RNN in processing of MI-EEG.

Index Terms - Brain computer interface; Long Short-Term Memory; Recurrent Neural Network; Discrete Wavelet Transform; motor imagery EEG;

I. INTRODUCTION

Brain-computer Interface (BCI) is a direct communication pathway between the brain and an external device, when the brain and peripheral nerve and muscle tissue pathway was blocked. It is an entirely new way of human-computer interaction [1]. With the speeding up of global ageing degree, old people are more likely to develop the stroke hemiplegia. Medical theory and practice prove that the brain is plasticity, actively participate in sports training is more conducive to the rehabilitation of patients [2,3]. The motor imagery and BCI based rehabilitation robot provides a new way of active rehabilitation for motor dysfunction, which has obvious advantages in improving the recovery of motor function [4]. BCI has become a research hotspot in the field of Rehabilitation robot.

Since Motor Imagery Electroencephalography (MI-EEG) is usually derived from scalp EEG, it has the characteristics of nonlinear, non-stationary and time-varying sensitivity. The key problem of BCI system is how to identify EEG quickly and accurately. Now, there are four main methods of MI-EEG

classifier: (1) Vector Machine Support [5] (SVM), which is used to find and construct the optimal hyper plane, is a statistical pattern recognition method. It has good generalization ability and nonlinear mapping ability, and has been commonly used in the field of EEG. (2) Naive Bayes (NB) [6] is a kind of probability statistical method. Because simple and easy to implement, it is often applied in the field of EEG, but it is not good for the EEG data with large noise. (3) Linear Discriminant Analysis [7] (LDA) is simple and feasible, and it is suitable for the analysis of linear signals, but it is not good for the nonlinear EEG signals. (4) Back Propagation [8] (BP) neural network has strong nonlinear mapping capability, but it is easy to fall into local optimum, so its classification effect is not very good. However, the EEG signal is a kind of time series signal, and the present methods have not considered the time sequence information of EEG, perhaps the resulting recognition accuracy is not very ideal.

The Recurrent Neural Network (RNN), and specifically a variant with Long Short-Term Memory (LSTM) [9], have recently emerged as an effective model in a wide variety of applications that involve sequential data. These include speech recognition [10], handwriting recognition and generation [11], video analysis [12] and image captioning [13]. However, RNN have been not widely used in the field of EEG. Petrosian, et al [14] first applied to the field of EEG by recurrent neural network (RNN) and wavelet transform. RNN has achieved good results in intracranial EEG, but the effect is not satisfied in scalp EEG. On the one hand, because the scalp EEG contains more interference than the intracranial EEG, the input of the RNN does not have a special signal preprocessing. On the other hand, the simple RNN network has the problems such as gradient explosion and gradient vanish. The LSTM RNN can not only solve this problem but also store the long time information. Meanwhile, the network has good generalization ability and more good generalization ability.

In this paper, a LSTM based RNN is integrated with the Discrete Wavelet Transform (DWT) to yield a novel recognition method, denoted as DWT-LSTM. Firstly, the average power spectrum method is employed to analyze the observed MI-EEG signals and determine the effective time segment. Then, DWT is applied to each channel of MI-EEG to extract the effective time-frequency characteristics. Finally, a LSTM Network was used as a classifier to recognize the MI-EEG signals. Many experiments are conducted on a publicly available dataset with 5-fold cross validation. The

experimental results showed that DWT-LSTM method can make full use of the time-frequency information of MI-EEG, as well as time sequence information, and can achieve better recognition performance.

The rest of the paper consists of 3 sections. Section 2 describes the basic theory and method, including DWT, LSTM based RNN and so on. Section 3 presents the experimental research. The data source, experiment results and discussions are given in this Section. Section 4 concludes the paper.

II. THE BASIC THEORY AND METHOD

A. The discrete wavelet transform

The wavelet transform was a new development of fourier transform and its coefficients could reflect the local information of the signal both in time and frequency domain. Thus, as a time-frequency analysis method, it had a wide application in biomedical signal processing, especially for non-stationary signal such as EEG.

The DWT could be thought as a multiresolution decomposition of a signal. This meant that it could decompose a signal into its components in different frequency bands. The Inverse DWT (IDWT) did exactly the opposite.

The DWT of a signal $f(t)$ was given as follows:

$$C_{j,k}(f, \varphi_{j,k}) = 2^{-j/2} \sum_{n=-\infty}^{\infty} f(t) \varphi(2^{-j}t - k) \quad j, k \in \mathbb{Z} \quad (1)$$

Where $\varphi_{j,k}(t) = 2^{-j/2} \varphi(2^{-j}t - k)$ was a wavelet sequence, $\varphi(t)$ was the basic wavelet, j and k were the frequency resolution and time of the transform respectively. In wavelet analysis, there were the low-frequency, high-scale component of the signal as the approximation A and the high-frequency, low-scale component as the detail D . The extending scheme showing the components of a multi-level analysis was depicted in Fig.1. There were several methods to combine and reproduce the original signal. And the DWT decomposition would produce a hierarchically structured decomposition. The choice of the hierarchy was dependent on the signal. At each level j , the approximation A_j and the detail D_j were built. We considered the original signal as the approximation A_0 at level 0 and L as the level.

The signal $f(t)$ was decomposed into its components in different frequency bands. $f(t)$ can be expressed as follows:

$$\begin{aligned} f(t) &= A_1 + D_1 \\ &= A_2 + D_2 + D_1 \\ &= A_3 + D_3 + D_2 + D_1 \\ &= A_4 + D_4 + D_3 + D_2 + D_1 \\ &= A_L + \sum_{j=1}^L D_j \end{aligned} \quad (2)$$

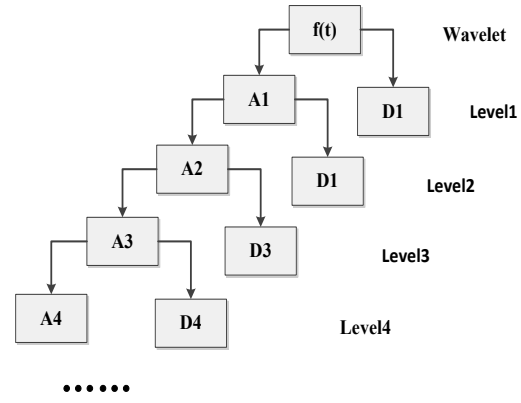


Fig 1. A hierarchically organized decomposition.

When the EEG was sampled with f_s , the corresponding bands of $A_L, D_L, D_{L-1} \dots D_1$ would be as follows:

$$\left[0, \frac{f_s}{2^{L+1}} \right], \left[\frac{f_s}{2^{L+1}}, \frac{f_s}{2^L} \right] \dots \left[\frac{f_s}{2^2}, \frac{f_s}{2} \right].$$

B. LSTM based recurrent network

The recurrent neural network is a strict superset of feedforward neural networks, augmented by the inclusion of recurrent edges that span adjacent time steps, introducing a notion of time to the model. The feedforward neural network is presented in Fig.2 and the recurrent neural network is depicted in Fig.3.

In RNN, the hidden layer function h is an elementwise application of a sigmoid function. However we have found that the LSTM architecture [15], which uses purpose-built memory cells to store information, is better at finding and exploiting

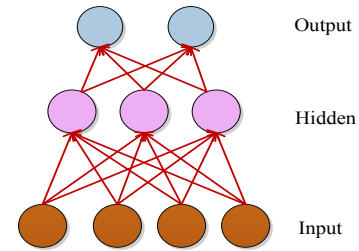


Fig 2. The framework of the feedforward neural network

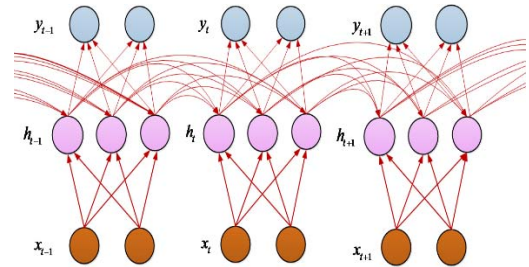


Fig 3. The framework of the recurrent neural network.

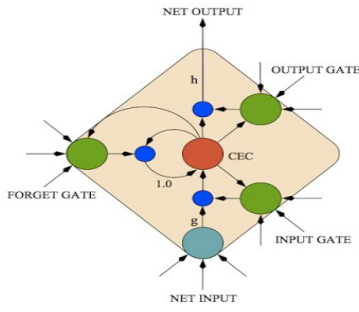


Fig 4. LSTM memory cell.

long range dependencies in the data. Fig. 4 illustrates a single LSTM memory cell. For the version of LSTM used in this paper [16], LSTM is implemented by the following composite function:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \quad (3)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \quad (4)$$

$$c_t = f_{t-1} + i_t \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \quad (5)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_t + b_o) \quad (6)$$

$$h_t = o_t \tanh(c_t) \quad (7)$$

Where σ is the logistic sigmoid function, i, f, o and c are respectively the input gate, forget gate, output gate, cell and cell input activation vectors, all of which are the same size as the hidden vector h . The weight matrix subscripts have the obvious meaning, for example W_{hi} is the hidden-input gate matrix, W_{xo} is the input-output gate matrix etc. The weight matrices from the cell to gate vectors (e.g. W_{ci}) are diagonal, so element m in each gate vector only receives input from element m of the cell vector. The bias terms (which are added to i, f, o and c) have been omitted for clarity.

The original LSTM algorithm used a custom designed approximate gradient calculation that allowed the weights to be updated after every timestep [15]. However the full gradient can instead be calculated with backpropagation through time [18], and the method is used in this paper. One difficulty when training LSTM with the full gradient is that the derivatives sometimes become excessively large, leading to numerical problems. To prevent this, all the experiments in this paper used clipped the derivative of the loss with respect to the network inputs to the LSTM layers (before the sigmoid and tanh functions are applied) [16].

III. EXPERIMENTAL RESEARCH

A. Experimental data

The experiment data were from Data set III of the “BCI Competition 2003” contest database. There are 140 training trails and 140 test trails in the dataset. The process of experimental data acquisition were shown in Fig.5,

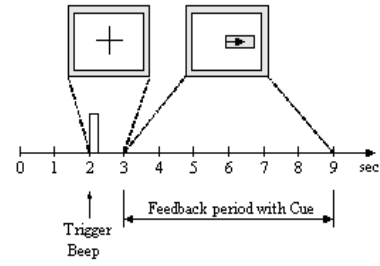


Fig 5. The scheduling of motor imagery experiment.

acquisitions were obtained every 9s, two minutes interval between experiments. The first 2s were quite, at $t=2s$, an acoustic stimulus indicated the beginning of the trial, and a cross “+” was displayed for 1s; then at $t=3s$, an arrow (left or right) was displayed for 1s; then at $t=3s$, an arrow (left or right) was displayed as cue. At the same time the subject was asked to move a bar into the direction of the cue. The recording was made using a G.tec amplifier and some Ag/AgCl electrodes. Three bipolar MI-EEG channels (anterior ‘+’, posterior ‘-’) were measured over C3, Cz and C4. The MI-EEG was sampled with 128Hz [19].

B. Time series analysis of MI-EEG

MI-EEG collected through the electrode cap was stored as the voltage amplitude, so the instantaneous power was calculated as follows:

$$P_i^j(t) = [x_i^j(t)]^2 \quad (8)$$

Where, $t = \{1, 2, \dots\}$ is the number of experiments, $i = 1, 2, \dots$ is the number of channel, $j = \{1, 2, \dots\}$ is the number of sample point. $x_i^j(t)$ is the data of the j^{th} sample point in the i^{th} trial, $P_i^j(t)$ is the instantaneous power of the j^{th} sample data in the i^{th} trial.

Assuming the average power of the j^{th} sample point data by N trials, the average power calculated in (9):

$$P_i^j = \frac{1}{N} \sum_{t=1}^N P_i^j(t) \quad (9)$$

When $N=140$, the average power during 0~9s was obtained for left-hand on electrode C3 or C4 respectively, namely

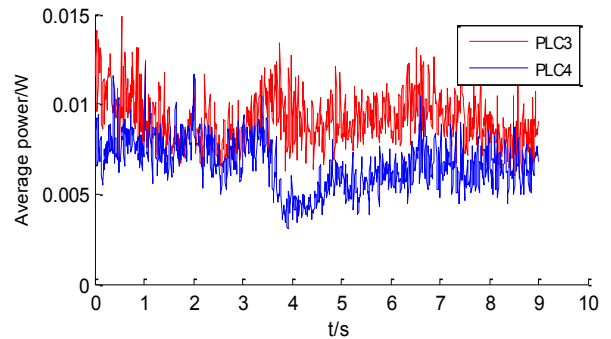


Fig 6. Variation of the average power of C3 and C4 with time for imagining left-hand movement.

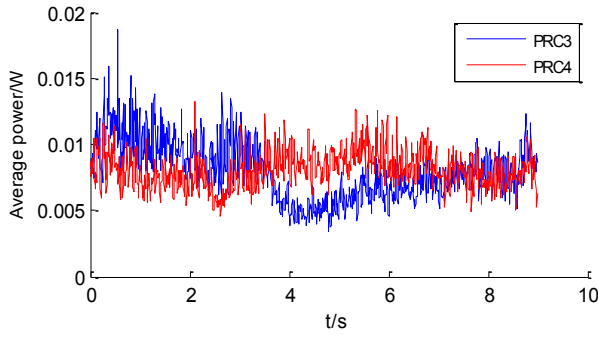


Fig 7. Variation of the average power of C3 and C4 with time for imagining right-hand movement.

PLC3, PLC4, and for right-hand on electrode C3 or C4: PRC3, PRC4. From Fig.6 and Fig.7, there was an obvious difference between left-imagery and right-imagery in 3.5~8s. Therefore, we selected the EEG signals within the 3.5s~8s segment for feature extraction.

C. Feature extraction with DWT

The average power of imagining left-right hand movements was decomposed into 6 levels by Daubechies 4-tap wavelet, and thus. The corresponding bands were followed by 0Hz~1Hz, 1Hz~2Hz, 2Hz~4Hz, 4Hz~8Hz, 8Hz~16Hz, 16Hz~32Hz, 32Hz~64Hz. The decomposed results were shown in Fig.8 and Fig.9, where L3A6 and L4A6 were A6 of left-imagery on C3 channel and C4 channel respectively, and R3A6, R4A6 were the counterpoint of right-imagery.

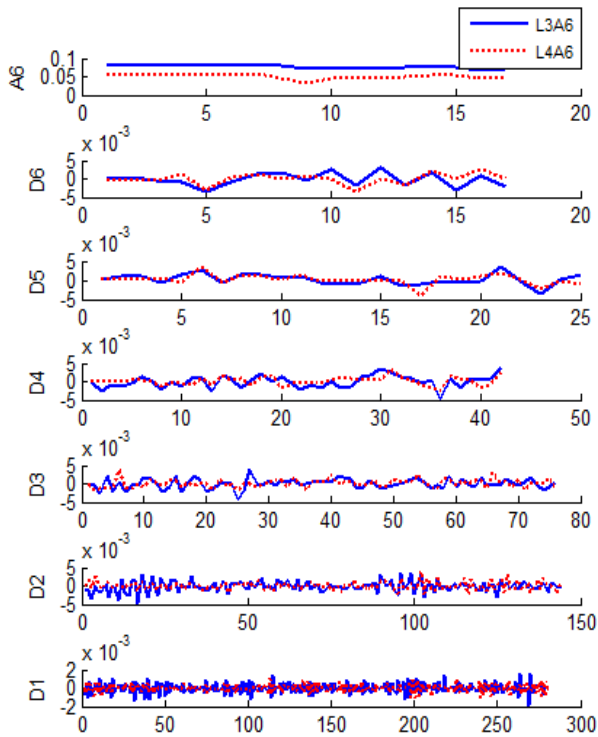


Fig 8. The wavelet decomposition of the C3 and C4 for imagining left-hand movement.

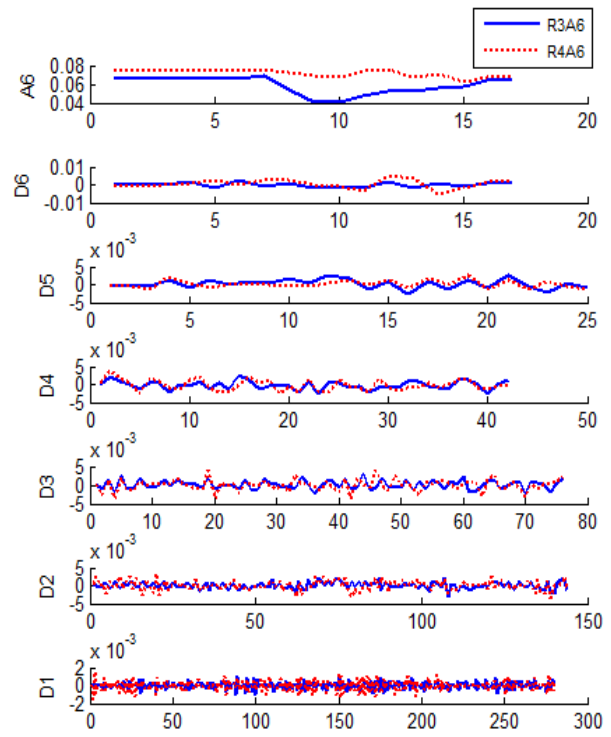


Fig 9. The wavelet decomposition of the C3 and C4 for imagining right-hand movement.

As shown in the Fig.8 and Fig.9, the C3 of the left hand L3A6 is higher than that of the left C4 on the scale 6. The R3A6 is lower than the C4 of the A6 on the right hand R4A6 at the scale 6. At the same time, we can see that the feature of MI-EEG extracted by DWT is still maintained the original sequence feature. LSTM recurrent networks classifier can make full use of its temporal features. So this paper chooses the approximation signal A6 as the characteristic of the scale 6. Assuming $C_3A_6 \in R^{1 \times 17}$ represents A_6 of Wavelet packet coefficients on C_3 channel, $C_4A_6 \in R^{1 \times 17}$ represents A_6 of Wavelet packet coefficients on C_4 channel, $C_ZA_6 \in R^{1 \times 17}$ represents A_6 of Wavelet packet coefficients on C_Z channel. The MI-EEG feature can be calculated in (10):

$$F = C_3A_6 + C_4A_6 + C_ZA_6 \quad (10)$$

Where, $F \in R^{3 \times 17}$ as the feature vector to distinguish left-hand motor imagery and right-hand motor imagery. The sign “+” represents serial feature fusion.

D. Classification based on a LSTM based recurrent network

LSTM based recurrent networks can store the time sequence information, the use of the time series function and good generalization ability. In this paper, a LSTM based recurrent network is used as the EEG classifier. Because of the feature of MI-EEG is $F \in R^{3 \times 17}$, we can get the length of MI-EEG features sequence is 17 and the features number of MI-EEG is 3 at the same time. Therefore, the neural network input layer selects 3 neurons, and the output layer is 1 neurons.

In order to objectively evaluate the classification results and obtain a reliable and stable model, the 5-fold cross validation is employed to calculate the classification accuracy. Some experiments are conducted to determine the neurons number of hidden layer in LSTM network, and the experiment results are shown in Table 1. So the best recognition accuracy is 93.30%, and the corresponding neurons number of hidden layer is 9. The structure of LSTM network will be applied in the follow-up studies.

TABLE I
THE EFFECT OF NEURONS NUMBER IN HIDDEN LAYER TO THE CLASSIFICATION ACCURACY

The neurons number of hidden layer	Accuracy
7	90.18%
8	91.52%
9	93.30%
10	92.41%

E. Comparison with other methods

Based on Data set III of the “BCI Competition 2003” contest database, the proposed method is compared with the top three of “BCI Competition 2003” [19] in many aspects, such as feature extraction, classifier and recognition rate. The experiment results are shown in Table 2. As can be seen from the table, the method of this paper has increased the recognition rate at least about 4%.

TABLE II
METHODS AND RESULTS OF THREE METHODS IN BCI CONTEST

Author	Feature extraction	Classification	Recognition rate
Christin Schafer	Wavelet	Bayes	89.29%
GAO Xiaorong	ERD	LDA	86.43%
Akash Narayana	AR	LDA	84.29%
Author in this paper	DWT	LSTM	93.30%

Furthermore, in order to verify the effectiveness of the LSTM based RNN classifier, the more experiments are conducted to compare with KNN, Bayes, LDA and SVM.

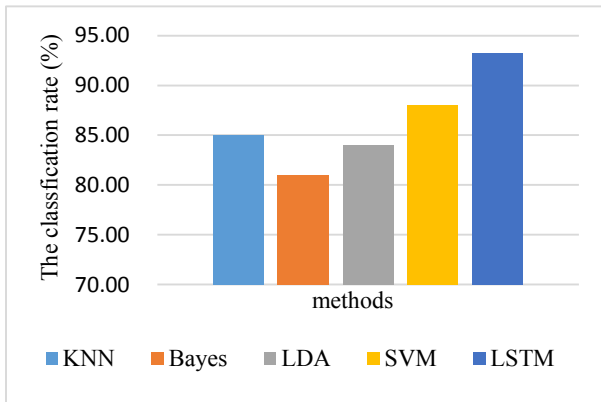


Fig 10. The comparison with other four classifiers.

Here, the same MI-EEG feature set $F \in R^{3*17}$ from subsection C in section III is used, the classification results are shown in Fig.10.

Based on the same data set, the proposed recognition method, i.e., WT+LSTM, is compared with other classical methods such as WP + LDA [20], WT + LDA [21], WPD + Fisher [22], WPD + SVM [23]. The results are shown in Fig.11.

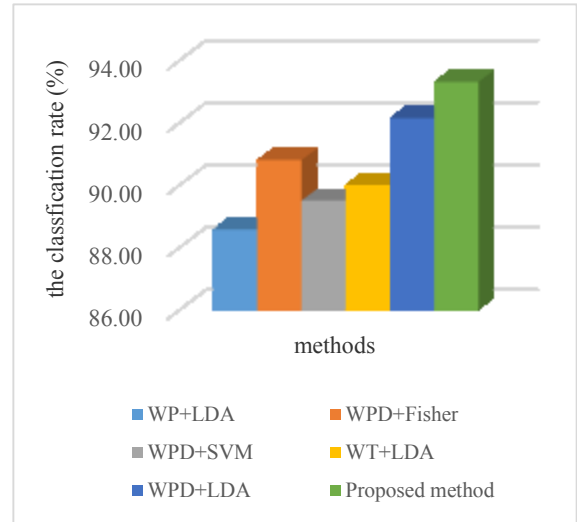


Fig 11. The comparison of multi recognition methods.

From Fig.11, we can see that the proposed method is better than the others and it can effectively improve the classification accuracy of MI-EEG. Therefore, DWT-LSTM is provided for the classification and application of MI-EEG.

IV. CONCLUSIONS

According to the time series of MI-EEG signals, a recognition method based on DWT and LSTM recurrent network is proposed, denoted as DWT-LSTM. Firstly, DWT is used to analyze the non-stationary time series of MI-EEG, and the features in the time and frequency domain are extracted, and it still remains the time sequence information of the original signal. Then, the time-frequency features are classified using LSTM recurrent network. The experimental results show that the DWT-LSTM method can make full use of the time domain and frequency domain information of MI-EEG, as well as its time sequence information. and it obtains a relatively higher recognition rate. It is noticeable that LSTM is introduced into the processing of MI-EEG, and this has important significance for the further research and practical application of MI-EEG in rehabilitation field.

ACKNOWLEDGMENT

This work is financially supported by the National Natural Science Foundation of China (No.81471770), the Natural Science Foundation of Beijing (No.7132021) and the Promotion Projects of Beijing University of Technology. The authors are grateful to the anonymous referee for carefully checking the details and for helpful comments that improved this paper.

REFERENCES

- [1] Wolpaw J R, Birbaumer N, Heetderks W J, et al. Brain-computer interface technology: a review of the first international meeting [J]. *IEEE transactions on rehabilitation engineering*, 2000, 8(2): 164-173.
- [2] Fiori M, Tinazzi M, Aglioti SM. Selective impairment of hand mental rotation in patients with focal hand destonia. *Brain*, 129(1), pp47-54, 2006
- [3] Ungerleider L G, Doyon J, Karni A. Imaging brain plasticity during motor skill learning [J]. *Neurobiology of learning and memory*, 2002, 78(3): 553-564.
- [4] Fu M J, Daly J J, Cavusoglu M C. Assessment of EEG event-related desynchronization in stroke survivors performing shoulder-elbow movements[C]//*Robotics and Automation*, 2006. ICRA 2006. Proceedings 2006 IEEE International Conference on. IEEE, 2006: 3158-3164.
- [5] Kim S H, Faloutsos C, Yang H J. Coercively adjusted auto regression model for forecasting in epilepsy EEG [J]. *Computational and mathematical methods in medicine*, 2013, 2013.
- [6] Machado J, Balbinot A. Executed Movement Using EEG Signals through a Naive Bayes Classifier [J]. *Micromachines*, 2014, 5(4): 1082-1105.
- [7] Subasi A, Gursoy M I. EEG signal classification using PCA, ICA, LDA and support vector machines [J]. *Expert Systems with Applications*, 2010, 37(12): 8659-8666.
- [8] Zhou Y, Li J, Zhou X. Study on EEG Password based on AR Power Spectrum and BP network [J]. *Journal of Convergence Information Technology*, 2012, 7(9).
- [9] Hochreiter S, Schmidhuber J. Long short-term memory [J]. *Neural computation*, 1997, 9(8): 1735-1780.
- [10] Graves, Alan, Abdel-rahman Mohamed, and Geoffrey Hinton. "Speech recognition with deep recurrent neural networks." *Acoustics, Speech and Signal Processing (ICASSP)*, 2013 IEEE International Conference on. IEEE, 2013.
- [11] Graves A. Generating sequences with recurrent neural networks [J]. *arXiv preprint arXiv:1308.0850*, 2013.
- [12] Donahue, Jeff, et al. "Long-term recurrent convolutional networks for visual recognition and description." *arXiv preprint arXiv: 1411.4389* (2014).
- [13] A. Karpathy and L. Fei-Fei. Deep visual-semantic alignments for generating image descriptions. *CVPR*, 2015.
- [14] Petrosian, Arthur, et al. "Recurrent neural network based prediction of epileptic seizures in intra-and extracranial EEG." *Neurocomputing* 30.1 (2000): 201-218.
- [15] Hochreiter, Sepp, and Jürgen Schmidhuber. "Long short-term memory." *Neural computation* 9.8 (1997): 1735-1780.
- [16] Graves, Alex. "Generating sequences with recurrent neural networks." *arXiv preprint arXiv: 1308.0850* (2013).
- [17] Graves, Alex, and Jürgen Schmidhuber. "Framewise phoneme classification with bidirectional LSTM and other neural network architectures." *Neural Networks* 18.5 (2005): 602-610.
- [18] Schlögl A, Lugger K, Pfurtscheller G. Using adaptive autoregressive parameters for a brain-computer-interface experiment. *The 19th Annual International Conference IEEE/EMBS*, Chicago, USA, 1997
- [19] Blankertz B, Müller K R, Curio G, et al. The BCI competition 2003: progress and perspectives in detection and discrimination of EEG single trials [J]. *Biomedical Engineering, IEEE Transactions on*, 2004, 51(6): 1044-1051.
- [20] Baoguo X, Aiguo S. EEG signal recognition method based on wavelet packet transform and clustering analysis [J]. *Chinese Journal of Scientific Instrument*, 2009, 1: 007.
- [21] Xu B, Song A, Wu J. Algorithm of imagined left-right hand movement classification based on wavelet transform and AR parameter model[C]//*Bioinformatics and Biomedical Engineering*, 2007. ICBBE 2007. The 1st International Conference on. IEEE, 2007: 539-542.
- [22] Ting W, Guozheng Y, Banghua Y. EEG feature extraction in brain computer interface based on wavelet packet decomposition [J]. *Chinese Journal of Scientific Instrument*, 2007, 28(12): 2230.
- [23] Yanjing W, Xiaoyan Q, Peng L, et al. Classification of motor imagery task based on wavelet packet entropy and support vector machines [J]. *Chinese Journal of Scientific Instrument*, 2010, 12: 2729-2735.